



Chapter 1

Introduction to Oracle DBMS

CT208H – Oracle Database Management System



Content

1. Oracle 11g Enterprise Edition
2. Database user types
3. SQL, PL/SQL, SQL*Plus and Oracle SQL Developer
4. Oracle versions and editions
5. Administration accounts & security



1. ORACLE 11G ENTERPRISE EDITION

Chapter 1. Introduction to Oracle DBMS



What is a database?

- A **database** is an organized **collection of structured data** stored electronically in a computer system
- Before the database system was invented, the **flat file** structure was commonly used to store data
 - When the number of rows in the flat file is **increased** e.g., million rows, it becomes **unmanageable**

```
first name, last name, phone
John, Doe, (408)-245-2345
Jane, Doe, (503)-234-2355
...
```



Relational database model

- Invented by Dr. Ted Codd in the 1970s
- Data is organized into **entities** and **attributes** instead of combining everything in a single structure
 - An **entity** (row/record) is a person, place, or thing
 - **Attributes** (field) describe the person, place, and thing
 - Similar entities are stored in a **table**
 - Related tables are stored in a **database**
 - **Relations** of entities are **described by data**





Relational Database Management Systems

- Database Management Systems based on **relational data model** (RDBMS in short)
- The most **popular** type of DBMS
- Some other notable ones:
 - Oracle (from Oracle Corporation)
 - DB2 (from IBM)
 - SQL Server (from Microsoft)
 - MySQL (open-source RDBMS, also from Oracle)



Oracle Database Management System

- One of the **most popular RDBMS**, developed by Oracle Corporation in 1977
- Oracle database **characteristics**:
 - **Cross-platform**: Windows, Unix, GNU/Linux
 - **ACID-compliant**: ensures data integrity and reliability
 - Oracle **networking** stack: applications from different platforms can communicate with each other
 - Commitment to **open technologies**: supports open GNU/Linux



Oracle Database Management System

- Important features:





Oracle Database Management System

- Availability:
 - Supported 24*7 availability of the database
 - Oracle Data Guard functionality: allows using of the secondary database as a copy of the primary database during any failure
- Security:
 - Unauthorized accesses are prevented
 - Oracle Advanced Security: offers Transparent Data Encryption (source level or after export) and Data Redaction (application level)



Oracle database management system

- Scalability:
 - Provides features like RAC (**Real Application Cluster**) and **Portability**, which makes an Oracle DB scalable based on usage
 - RAC provides capabilities such as rolling instance migrations, performing upgrades, maintaining application continuity, etc.
- Performance: Provides many performance optimization tools, such as **Oracle Advanced Compression**, and **Oracle Database In-Memory**,... to improve system performance



Oracle database management system

- Analytics:
 - OLAP (**Oracle Analytic Processing**): provides complicated analytical calculations on business data
 - **Oracle Advanced Analytics**: assists customers in determining **predictive business models** through data and text mining, as well as **statistical data** computation
- Management: **Oracle Multitenant** combines a single container database with many pluggable databases in a consolidated design



Oracle database management system

- Some other features:
 - **Logical** data structure: users can interact with the database without knowing where the data is stored **physically**
 - **Partitioning**: a large table can be divided into **different pieces**, and pieces can be stored **across storage devices**
 - **Memory caching**: allows scaling up a **very large database** that still can perform at a **high speed**
 - **Portability**: Oracle database can be ported on all **different platforms** than any of its competitors



Oracle database management system

- Some other features:
 - Backup and recovery:
 - Ensures the **integrity** of the data in case of **system failure**
 - The Recovery Manager (RMAN) tool allows DBA to perform cold, hot, and incremental database backups and point-in-time recovery
 - Clustering: enables **high availability** that enables the system to be up and running **without interruption** of services in case one or more servers in a **cluster fails**



Oracle database management system

- Disadvantages:
 - Complexity: Oracle is not recommended for use when the users are **not technically savvy** and have limited technical skills required to deal with the Oracle Database
 - Cost: the price of Oracle products is **very high** in comparison to other databases
 - Difficult to manage: Oracle databases are often much **more complex** and difficult in terms of the management of certain activities



2. ORACLE DATABASE USER TYPES

Chapter 1. Introduction to Oracle DBMS



Types of Oracle users

- Database Administrators
- Security Officers
- Application Developers
- Application Administrators
- Database Users
- Network Administrators



Types of Oracle users

- Database administrators:
 - Oracle DBs are **large** and have **many users**
 - Main responsibilities:
 - **Installing** and **upgrading** the Oracle Server and application tools
 - **Managing DBs** (storage, tablespaces, DB objects)
 - **Enrolling users** and maintaining system **security**
 - Monitoring and optimizing the **performance** of the database
 - Planning for **backup** and **recovery** of database information



Types of Oracle users

- Application developers:
 - Design and implement **database applications**
 - Main responsibilities:
 - Designing the **database structure** for an application
 - **Tuning** the application during development
 - Establishing an application's **security** measures during development
- Application administrators: administrate a particular application



Types of Oracle users

- Security officers: concerned with **enrolling users**, controlling and monitoring **user access** to the database, and maintaining **system security**
- Database users: interact with the database **via applications** or utilities
 - entering, modifying, and deleting data, where permitted
 - generating reports of data
- Network administrators: administering Oracle **networking products**, such as Net8



3. SQL, PL/SQL, SQL*PLUS AND SQL DEVELOPER

Chapter 1. Introduction to Oracle DBMS



- A **set-based declarative language** that provides an **interface** to an **RDBMS** such as Oracle Database
- A **nonprocedural** that describes **what** should be done, **not how** to derive it
- SQL is the **ANSI standard language** for relational databases.
- **Oracle SQL** is an implementation of the **ANSI standard** and supports numerous features that **extend beyond standard SQL**



- A procedural extension to Oracle SQL
- Enables to use all of the Oracle Database SQL statements, functions, and data types
- Used to control the flow of SQL programs, create and use variables, and write error-handling procedures
- A primary benefit of PL/SQL is the ability to store application logic in the database itself
=> Built-in functionalities can be deployed anywhere



SQL*Plus

- A **command-line** interactive user interface and **batch query tool** that is installed with every Oracle Database installation
- Has its **own commands** and environment, and it **provides access** to the Oracle Database
 - Format, perform calculations, store, and print from query results
 - Examine table and object definitions
 - Develop and run batch scripts
 - Perform database administration



SQL*Plus

The screenshot shows a standard Windows-style application window titled "Oracle SQL*Plus". The menu bar includes "File", "Edit", "Search", "Options", and "Help". The main text area displays the following information: the SQL*Plus release version (10.1.0.4.2), the production date and time (Tue Jan 8 10:44:22 2013), the copyright notice (© 1982, 2005, Oracle), and the connection details (Oracle Database 11g Express Edition Release 11.2.0.2.0). The prompt "SQL>" is followed by a vertical cursor bar.

```
Oracle SQL*Plus
File Edit Search Options Help

SQL*Plus: Release 10.1.0.4.2 - Production on Tue Jan 8 10:44:22 2013

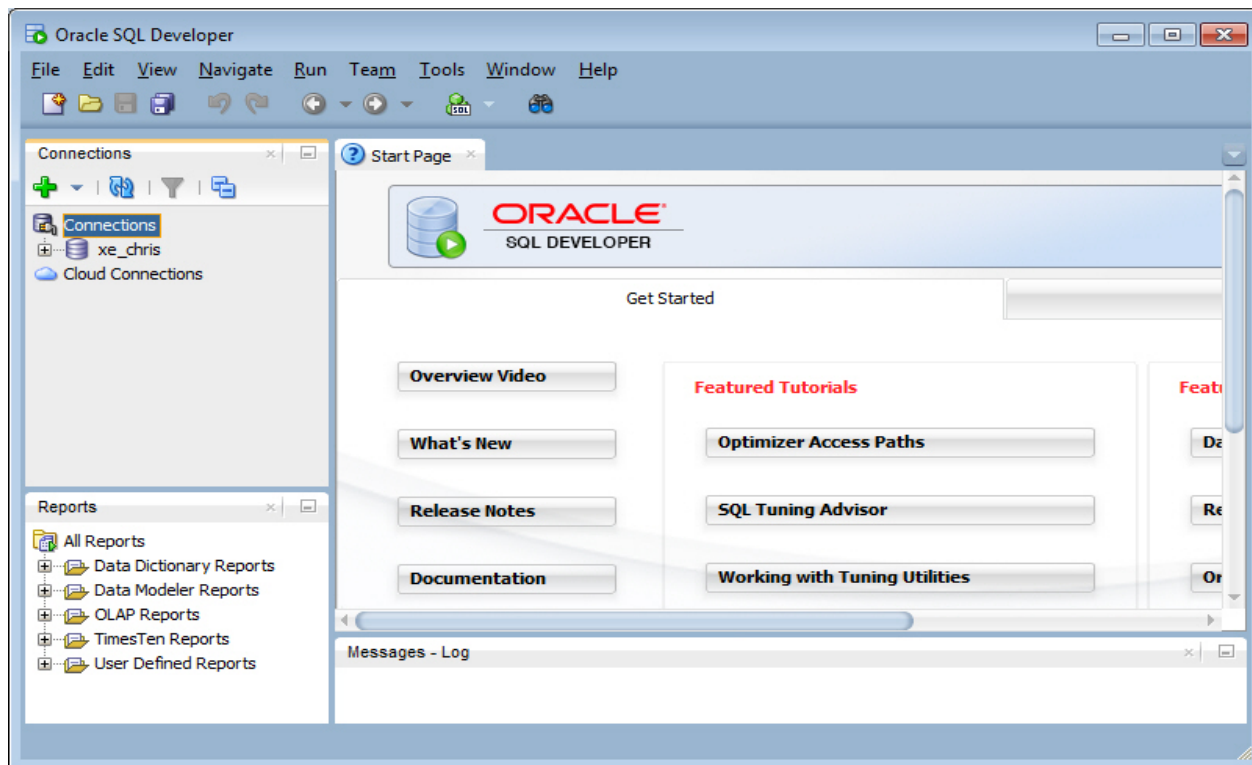
Copyright (c) 1982, 2005, Oracle. All rights reserved.

Connected to:
Oracle Database 11g Express Edition Release 11.2.0.2.0 - Production

SQL> |
```




Oracle SQL Developer





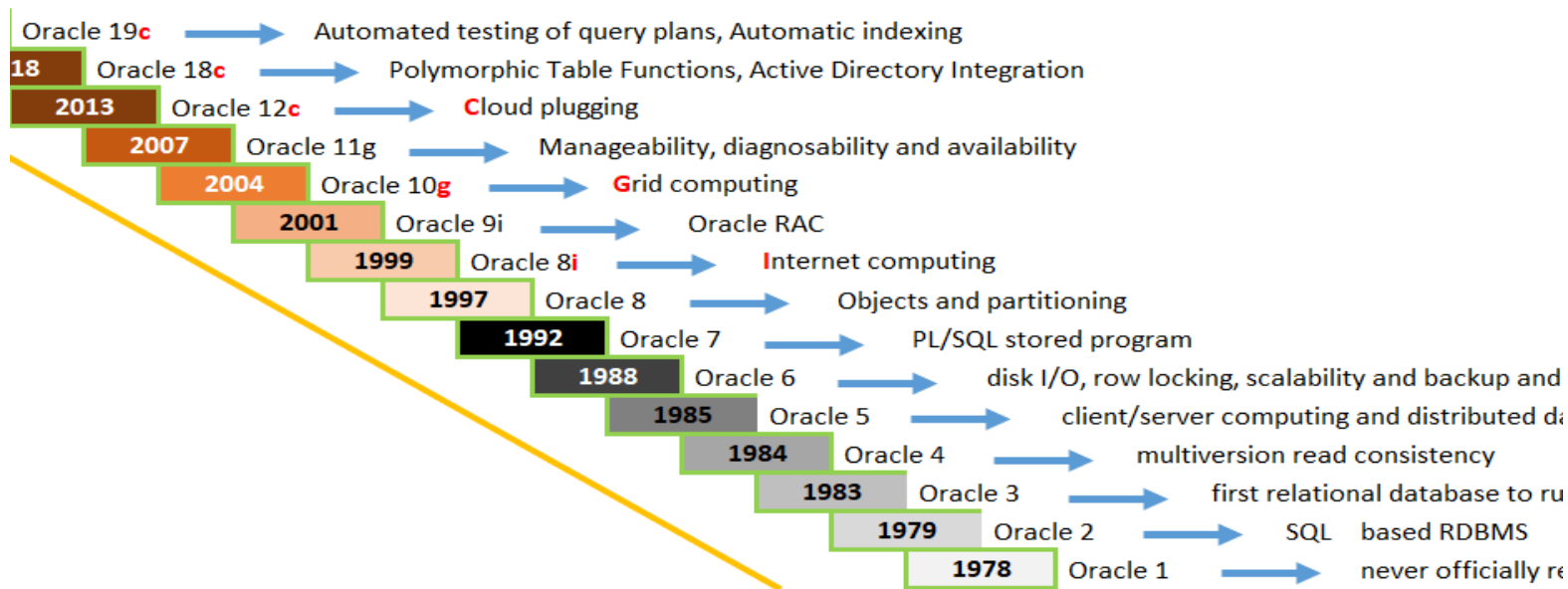
4. ORACLE VERSIONS AND EDITIONS

Chapter 1. Introduction to Oracle DBMS



Oracle Database versions

History of Oracle Database Versions





Oracle Database 11g editions

- Enterprise Edition:
 - Provides the performance, availability, scalability, and security required for **mission-critical applications** such as high-volume Online Transaction Processing (OLTP) applications, query-intensive data warehouses, and demanding Internet applications
 - Contains **all of the components** of the Oracle Database, and can be further enhanced with the purchase of the options and packs described



Oracle database 11g editions

- Standard Edition One:
 - Delivers unprecedented ease of use, power, and performance for workgroup, department-level, and Web applications
 - From **single-server** environments for **small business** to highly **distributed** branch environments
- Standard Edition:
 - Unprecedented ease of use, power, and performance of Standard Edition One
 - Supports **larger machines** and **clustering** of services with **Oracle Real Application Clusters** (Oracle RAC)



Oracle database 11g editions

- Personal Edition:
 - Supports single-user development and deployment environments that require full compatibility with all other editions
 - Includes all of the components and options that are included with Enterprise Edition, with the exception of the Oracle Real Application Clusters option and the Management Packs
 - Available on Windows and Linux platforms only



Oracle database 11g editions

- Express Edition:
 - An **entry-level** edition of Oracle Database
 - Quick to download, simple to install and manage, and is **free** to develop, deploy, and distribute
 - Can be installed on any size machine with any number of CPUs, stores up to **11GB of user data**, uses up to **1 GB of memory** and only **one CPU** on the host machine



5. ADMINISTRATION ACCOUNTS & SECURITY

Chapter 1. Introduction to Oracle DBMS



DB administrator security and privileges

- **Administrative tasks** in Oracle need **extra privileges** both **within the database** and possibly in the **operating system** (e.g., for software installation, executing OS commands, etc.)
- Access to a database administrator account should be **tightly controlled**
- Two user accounts are **automatically created** with the database and granted the DBA role: **SYS** and **SYSTEM**



The SYS account

- Automatically created and granted the **DBA role**
- All of the base tables and views for the database's **data dictionary** are stored in the schema SYS
- Tables in the **SYS schema** are manipulated **only by Oracle**
- Most database users should never be able to connect using the SYS account
- **DBA role**: contains all DB system privileges, and thus, it should only be granted to fully functional DB administrators



The SYSTEM account

- Automatically created and granted **all system privileges**
- The SYSTEM username creates **additional tables and views** that display administrative information and internal tables and views used by Oracle tools
- Never create tables of interest to individual users in the SYSTEM schema



DB administrator authentication

- DB administrators must often perform administrative operations such as shutting down or starting up a database
=> need a more secure authentication scheme than normal users
- Authentication methods:
 - Operating system authentication
 - Network authentication
 - Oracle database authentication
- Oracle allows using all methods within the same DB instance



Operating system authentication

- Benefits:
 - Once authenticated by the operating system, users can connect to Oracle more conveniently **without specifying a username or password**
 - With control over user authentication centralized in the operating system, **Oracle need not store or manage user passwords**, though it still maintains usernames in the database
 - **Audit trails** in the database and operating system use **the same usernames**



Network authentication

- Authentication capabilities at the network layer are handled by the **SSL protocol** or by **third-party services** such as:
 - DCE authentication
 - Kerberos authentication
 - Public Key Infrastructure-Based authentication
 - Authentication with RADIUS
 - Directory-based services



Oracle database authentication

- Oracle can authenticate users attempting to connect to a database by using information **stored in that database**
- To set up Oracle to use database authentication:
 - Creating each user with an associated password that must be supplied when the user attempts to establish a connection
 - Users' passwords are **stored in the data dictionary** in an **encrypted format** to prevent unauthorized alteration



QUESTIONS?

Chapter 1. Introduction to Oracle DBMS